DATAQUEST · THEME: EDUCATION

The Missing List

More than 40% of South African learners who start Grade 1 never finish matric. **Nobody publishes which schools.**

7,681	842	212,893	0.694
secondary schools scored for 2025	on the priority tier	learners inside those schools	test AUC, on years never seen

A school-level dropout risk register for South Africa, built entirely from open government data.

github.com/Mandla78/lp-dev-dataquest

The number is known. The list is not.

40 of every 100 children who walk into a Grade 1 classroom will not walk out of a Grade 12 one. That figure is published, widely quoted and not in dispute.

The problem is not that the number is unknown. It is that **it cannot be pointed at anything.**

Nobody can deploy a teacher to a percentage. An organisation with one retention programme and funding for twelve schools cannot use a national average to choose the twelve.

Learners are counted after they are gone, in aggregate, once a year.

Every year that list does not exist,

help goes somewhere —

just not necessarily where the loss is worst.

Four questions stand in the way. We answer three.

None of them currently has an answer anywhere in the public record.

Which provinces are losing the most?	8 of 9, ranked by measured risk	ANSWERED
Which districts inside those provinces?	72, ranked within their province	ANSWERED
Which schools inside those districts?	7,681, scored individually	ANSWERED
Which grades inside those schools?	Needs per-grade enrolment — held under permission	NOT YET

On the fourth we are deliberate. We can say a school is losing its Grade 8–12 pipeline. We cannot yet say whether the loss sits in Grade 10 or Grade 11. That is a form, not a rebuild — the same pipeline answers it the day the permission arrives.

THE CHALLENGE QUESTION

The data is broken, and nothing says so

Seven detectors over 572,607 records across two datasets.

572,607	19	13	0
records scanned	discrepancies found	severe enough to change a model	raise an error anywhere

2017 loses 1.44 million learners nationally; 2018 gains 1.31 million back.

Province codes are unstable — KZ/KZN, GT/GP — turning a country of nine provinces into a dataset of eleven.

Gauteng's 2023 file has its key column headed "G". Three thousand schools vanish.

724 schools sit at latitude zero — a point in the Atlantic Ocean.

Phase appears in 21 spellings, quintile in 17, the fee flag in 11.

8,792 schools change their recorded phase between years.

Not one of them raises an error. The files open, the row counts look reasonable, the totals look plausible. A crash gets noticed within the hour; a province that quietly becomes two survives into a published finding.

THE SAME STANDARD, APPLIED TO US

We audited our own work, and it failed too

Our first PDF parser trusted each report's header as the layout for the whole file. The reports do not work that way.

It produced 14,005 records claiming more learners were advanced into matric than sat the exam, and silently destroyed the pass-rate column.

So we removed pass rate from the model and wrote a comment explaining why. **The comment was wrong. The evidence behind it was our own bug.**

After the fix

pass rate 0% → 88.4%

coverage of the matric column

And the model

AUC 0.654 → 0.694

pass rate is now its strongest variable

The fix is verified by arithmetic the source itself prints: **where a report gives both the achieved count and the pass rate, the rate must equal achieved ÷ wrote. It holds on 82,979 of 82,979 checkable records.**

DATA INCLUSION

Two public datasets, joined on one key

Published openly by the Department of Basic Education. No login, no licence, no NDA, no purchase.

EMIS School Masterlists

478,761 records · 29,655 schools · 2007–2025 · all 9 provinces

Enrolment, educators, quintile, fee status, phase, district, coordinates. Built from 140 provincial spreadsheets.

education.gov.za › Programmes › EMIS Downloads

NSC School Performance Reports

93,846 records · 8,093 schools · 2008–2024

Candidates who wrote, achieved, the pass rate, and learners progressed without meeting requirements. Extracted by us from 13 PDFs — the department publishes no machine-readable version.

education.gov.za › Resources › Reports

Joined on the EMIS number, normalised to bare digits first — it is published both with and without a leading zero, and getting it wrong fails the join silently rather than loudly.

Matric evidence is carried forward only, never backward, with an age marker so a stale figure can be discounted. Carrying it backward would be reading the future.

Raw source files are committed alongside the code, so the extraction can be checked rather than taken on trust.

THE SOLUTION

What the machine learning actually does

Two supervised models were trained. The choice between them is the most consequential decision in the project.

Supervised learning

Learns from 61,485 historical school-years where the outcome is already known. Nothing is rule-based — the weights are fitted from data, not chosen by us.

Feature engineering

Raw registers become 31 candidate variables: enrolment trajectory, volatility, share of own peak, learners per educator, what neighbouring schools do, matric throughput. 15 survive into the model.

Weight of Evidence

Each variable is binned and re-expressed as the log-odds of loss it carries — letting a linear model capture non-linear risk and treat missing data as its own informative category.

Calibrated probabilities

Isotonic regression maps scores onto real observed rates, so a school at 0.30 loses learners about 30% of the time. Ranking gives order; calibration tells a planner how far their budget reaches.

Split by year, never at random. Trained on 61,485 school-years (2008–2016), tuned on 15,357 (2018–2019), measured once on 23,043 (2021–2023). A school recurs across years, so a random split would train and test on the same school.

RESULTS

Measured once, on years the model never saw

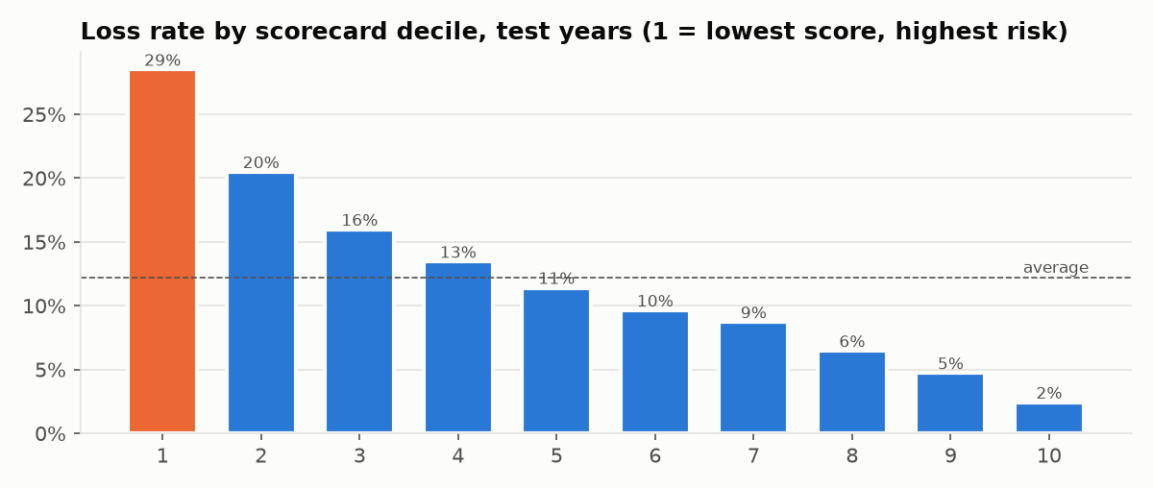
2021–2023. No refitting, no threshold tuning after the fact.

0.694
AUC on unseen years

0.388
Gini

0.281
KS statistic

2.34×
lift in the worst tenth



Worst tenth loses at 28.5% against 2.43% in the best tenth — the schools it ranks worst really are the ones losing learners.

Recall 53% at the threshold. A missed school gets nothing; a false alarm gets a visit. Those costs are not symmetric.

Accuracy is our weakest number and we say so. At 71% it is below the 88% you score by predicting nobody ever loses learners — a model that identifies no one. That is why we lead with lift.

THE DECISION WE HAD TO DEFEND

A better-ranking model exists. It is not what we ship.

SHIPPED

Logistic Regression Scorecard

AUC 0.6939 · lift 2.34×

Every score breaks into visible points. A school can be told it lost 37 points for its matric pass rate.

BENCHMARK, NOT SHIPPED

Calibrated Random Forest

AUC 0.7161 · lift 2.46×

Ranks better and cannot tell a school why it was flagged. Trained on the identical frame and split.

The gap is 0.022 AUC, and we publish it. This output directs public money. Whoever must justify the spend has to explain it line by line — to a treasury, a community, or a school that disagrees with its ranking. That is worth more than two hundredths of a ranking statistic, and stating the price makes the choice defensible rather than merely asserted.

We also measured the mistakes that inflate a model silently. **A random split lifts the tree from 0.692 to 0.731 — it has learned to recognise schools, not predict them.**

THE PRODUCT

A ranked list, and a reason beside every school

A planner filters to their province and sees what to work first — a morning's decision instead of a year's guesswork.

RANK 1 OF 7,681 · Vhembe East, Limpopo

Ntevhedzeni

111 learners **48% risk of losing learners faster than its province by 2027**

Against a national base rate of 12.2%.

WHY THIS SCHOOL

- matric pass rate: below 42.8 **(-37 pts)**
- enrolment volatility: 0.21 or more **(-17 pts)**
- school size (learners): below 225 **(-13 pts)**

That is the whole score. A school flagged for a pass-rate reason needs different help from one flagged for a staffing reason.

Coverage, stated plainly: 8 provinces, not 9. The Northern Cape published a Word template where its 2025 masterlist should be — a fault our own audit found, not one we introduced. The model covers all nine; the register covers the eight that published.

SYSTEM SECURITY

It holds nothing worth stealing — by design

Confidentiality is the easy half. The half that needs controls is integrity.

0	School	0	Public
personal records held	smallest unit of analysis	executable model artefacts	every source — nothing to steal

No personal information

Learner records are protected under POPIA and a hackathon team has no lawful basis to process them. A tool that names children creates a register of vulnerable minors.

Input validation at the boundary

Every download is checked against its format's magic bytes before it is written. A portal returning an HTML error page under a spreadsheet's filename is routine, and is rejected.

Nothing executable ships

The model is model.json — inert data. No pickle, no joblib, no eval. Loading a pickled model executes what is inside it.

The audit is an integrity control

Seven detectors already catch corrupted, truncated or altered source files. They caught our own parser producing 14,005 impossible records.

The gap we have not closed: dependencies are minimum-pinned but not hash-locked. A departmental deployment needs a lockfile and an internal mirror. A security section with no gap in it has not been written honestly.

BUSINESS MODEL

R612,000 a year, against what it prevents

A tool for the people who allocate support — never sold to the schools that need it.

R612k	R49	R0	4 provinces
to run for a year	per school scored	data acquisition — all sources public	break-even, at R150k each

WHAT THE PROBLEM COSTS

212,893 learners sit in the 842 schools the register flags as priority.

90 learners is the median two-year drop at a flagged school that does lose them — measured across 2,807 such cases, not assumed.

~21,600 learners is what that comes to across the 28.5% of the priority tier that loses.

WHO PAYS

Year 1 grant-funded pilot, two provinces

Year 2 four provinces at R150k, NGO tier at R25k

Year 3 nine provinces, national rollout

Adding a province is a filter, not a project.

The state has already spent about R200,000 educating one learner by Grade 10. **This costs less than three of them.**

The argument is not that the tool is cheap. It is that it is negligible against what is already being spent on the learners it exists to keep in school.

LIMITATIONS

What we are honest about

Every one of these is on the live application, not held back for the questions.

It measures enrolment decline, not dropout. A learner who transfers to a better school and one who leaves education entirely look identical in a headcount.

It is a triage list, not a verdict. At 0.694 AUC the ranking is worth working from the top of, and not good enough to be believed about any single school without a visit.

Population stability is a real caveat. PSI is 0.44 from training to the 2025 register, above the level at which a scorecard is normally rebuilt. Seventeen years separate them. It is the strongest argument for annual retraining.

The raw masterlists are incomplete in the repository. Four of the 140 spreadsheets remain; the built tables were produced from the full set and are committed in full. The README says so rather than implying otherwise.

A better model exists and is not shipped. The forest reaches 0.7161. We report the gap rather than hide it.

> A model that directs public money should be re-validated annually and withdrawn if it stops performing. **A model nobody re-checks is a governance risk long before it is a technical one.**

THE MISSING LIST

Open government data, a list a province can act on this week, and a reason next to every school.

7,681
schools scored

842
priority tier

19
data discrepancies found

0.694
AUC on unseen years

github.com/Mandla78/lp-dev-dataquest